

Full Project Report - Robotic Arm VLA

Text-Conditioned Robotic Manipulation via Vision-Language-Action Policies

Physical AI portfolio report for Grid Dynamics review

Stack: Python 3.11 / PyTorch / MuJoCo / gym-aloha / Hugging Face LeRobot

Release: v1.0.0

Repository: github.com/akobenhene/robotic-arm-vla

Framework source: <https://www.figma.com/board/CmvFbnixCtXsehIEUMbEnZ>

1. Abstract

This report presents an end-to-end Physical AI system for tabletop robotic manipulation. A bimanual Aloha robot in MuJoCo observes RGB imagery and proprioceptive state, optionally conditioned on natural language, and outputs continuous motor commands in a closed loop. Two LeRobot policies are compared: ACT, which solves TransferCube, and SmolVLA, which demonstrably changes actions when the prompt changes. Multi-seed evaluation yields 50% ACT success on CPU (25/50). The delivery includes CI, release artifacts, finetune and SO-100 docs, and Figma framework diagrams embedded below.

2. Introduction

Industrial automation is moving from fixed trajectories toward perception-driven, learning-based control. Clients ask for systems that see with cameras, optionally follow language work orders, validate safely in simulation, and improve from demonstrations. This project implements that loop on Aloha TransferCube: pick a red cube with the right arm and transfer it so the left arm holds it off the table (success when sparse reward reaches 4).

Contributions: closed-loop visuomotor control; multimodal VLA interface; quantitative multi-seed evaluation; comparative ablation narrative; and production-minded packaging suitable for Forward Deployed Engineering review.

3. Relevance for Grid Dynamics

Grid Dynamics invests in Physical AI / GAIN, digital twins, IoT and edge robotics, NVIDIA partnerships, and Forward Deployed Engineering. This project is a portable demonstration of the same delivery pattern: simulate, measure KPIs, refine imitation or VLA models, package for stakeholders, then plan hardware transfer.

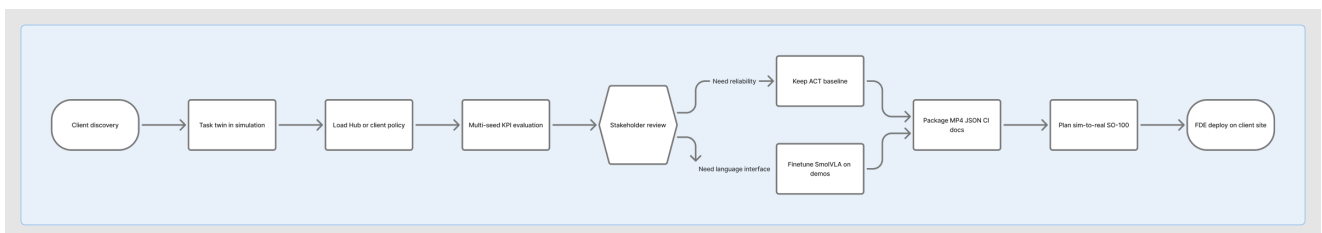


Figure 2. Grid Dynamics GAIN delivery loop (Figma).

3.1 Capability alignment

Grid Dynamics theme	Evidence in this project
Robotic manipulation	Closed-loop ACT on bimanual TransferCube
Validate before hardware	MuJoCo twin + 50-seed success metrics

Grid Dynamics theme	Evidence in this project
Imitation / foundation refine	Hub load + SmolVLA finetune playbook
Instruction-driven workflows	Language-conditioned SmolVLA ablation
Production engineering	CI smoke, release, PDF/MP4 artifacts
FDE ownership model	Env adapter to policy to KPI to docs

3.2 Client use

- PoC sprint: matched sim task, Hub policy, multi-seed eval in days.
- Stakeholder pack: MP4 for operators, JSON rates for CTOs.
- Modular policy swap: ACT today, finetuned SmolVLA tomorrow.
- Soft-sensing cross-sell: same multimodal stack for vision QC.

4. System design

The control loop is closed: each action changes the next image and state. RoboticsEnvWrapper converts gym-aloha observations into LeRobot batches. policy.py exposes ACT, SmolVLA, and a Mock backend behind predict().

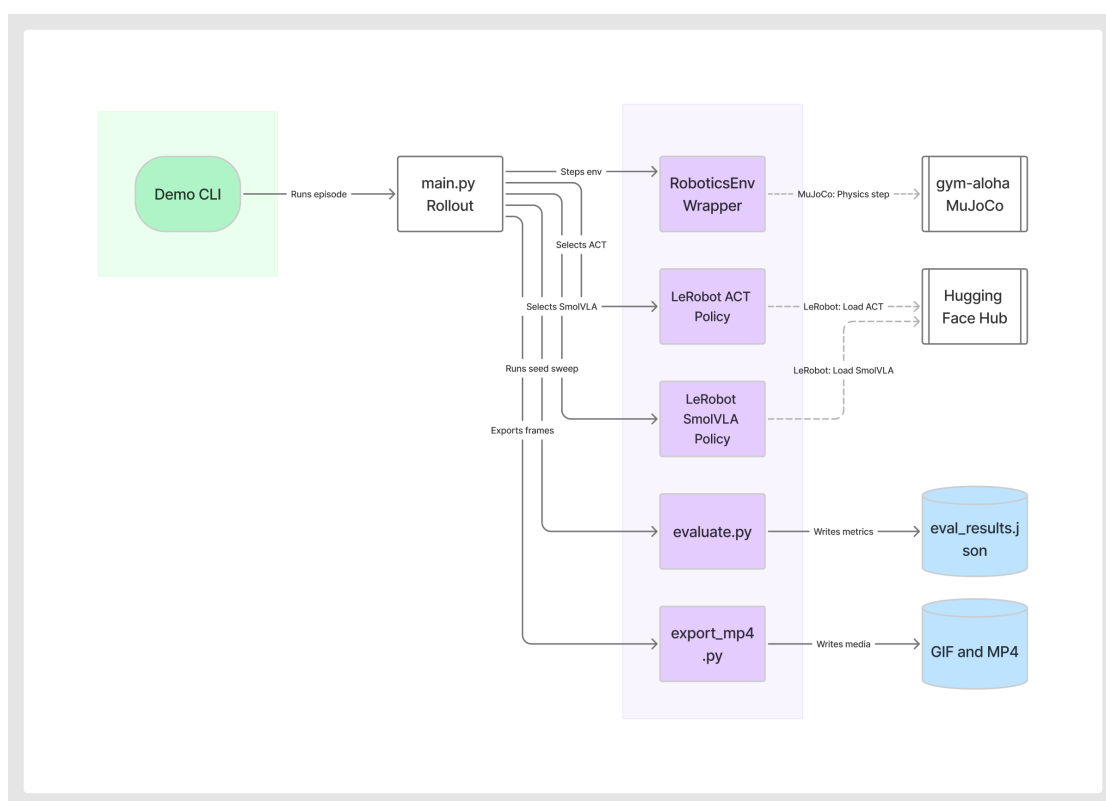


Figure 1. Physical AI system architecture (Figma).

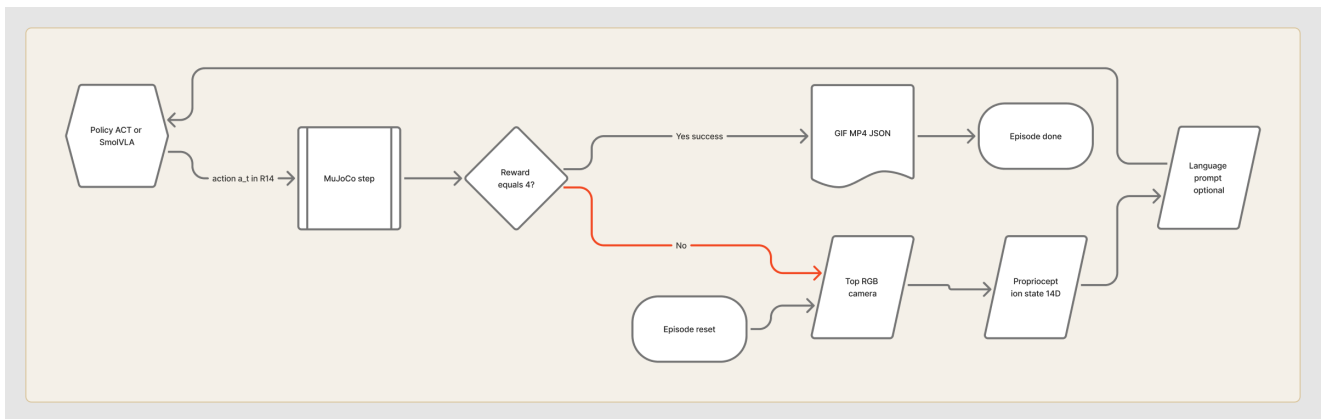


Figure 3. Closed-loop visuomotor control framework (Figma).

5. Task and reward ladder

TransferCube uses a sparse staged reward. Industrial KPIs are often similarly sparse (cycle complete / fail), so staged debugging matters in the field.

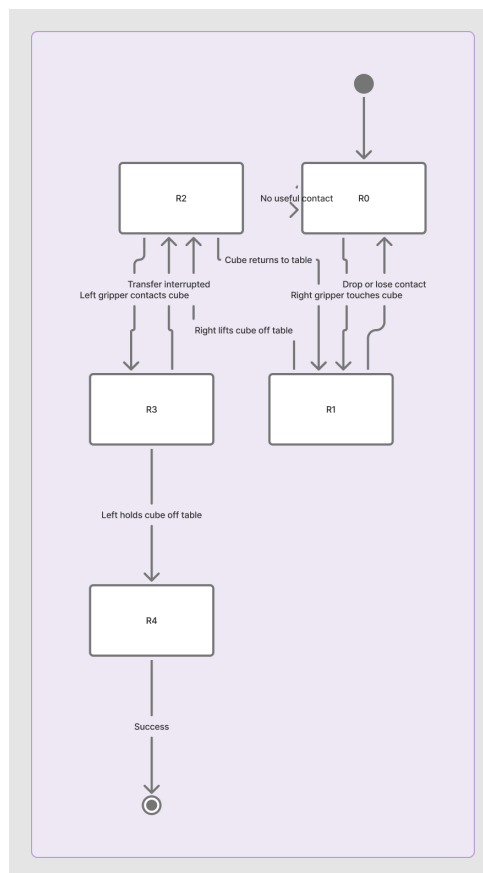


Figure 6. TransferCube sparse-reward state machine (Figma).

Reward	Meaning
0	No useful contact

Reward	Meaning
1	Right gripper touches cube
2	Right lifts cube off table
3	Left gripper contacts cube
4	Left holds cube off table (SUCCESS)

6. Evaluation framework

Task success and language sensitivity are measured separately. ACT is the task expert; SmolVLA proves the instruction interface. Conflating the two would mislead stakeholders.

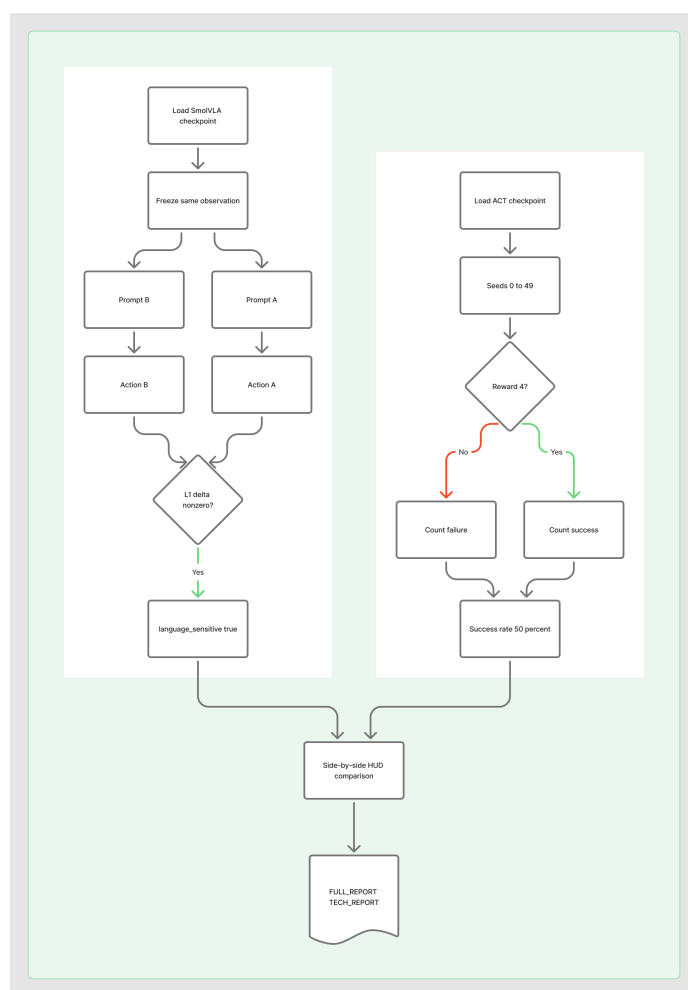


Figure 4. Evaluation and ablation framework (Figma).

6.1 Results

Metric	Value
ACT seeds evaluated	0-49 (50 episodes)
ACT success rate (CPU)	25/50 = 50.0%
Hero seed	36
Hub GPU reference	~83% / 500 episodes

Metric	Value
SmolVLA language sensitive	true (L1 action delta ~0.03)
SmolVLA full-task success	Not reliable on this checkpoint

7. Soft sensing and industrial estimation

AI soft sensors infer hard-to-measure quantities from cheap sensors and models. This robotics stack is the control counterpart: infer next actions from RGB and proprioception under physics constraints, then score with a KPI.

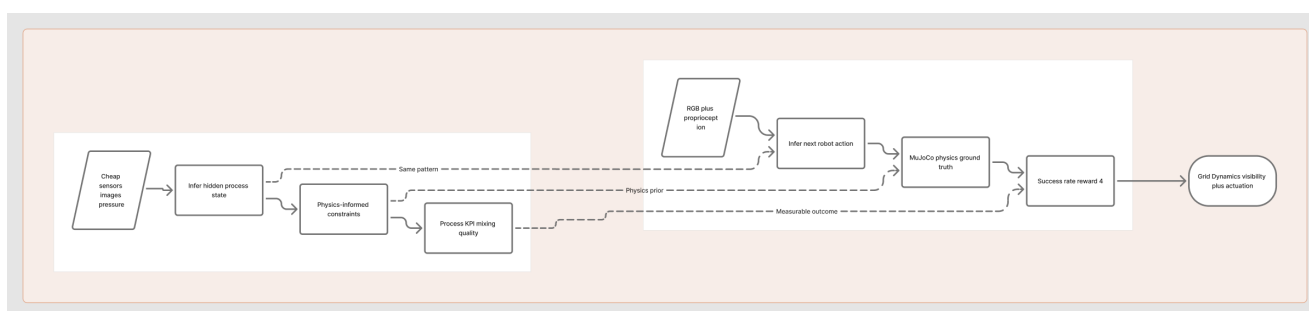


Figure 5. Soft-sensor research mapped to Physical AI (Figma).

8. Engineering lessons

- Zero-action warm-up after reset caused distribution shift and killed ACT success.
- Hub normalize buffers must be applied explicitly under LeRobot 0.4.x.
- Embodiment must match pretrained weights (Aloha, not FetchReach).
- Language sensitivity is not the same claim as task competence.
- Artifacts (GIF/MP4 + JSON + seeds) beat anecdotal demos.

9. Limitations and next steps

Limitation	Next step
SmolVLA weak on full transfer	GPU finetune on human TransferCube data
CPU eval below Hub 83%	GPU eval; more seeds
Sim only	SO-100 record, finetune, deploy path
Single task	Add insertion / pick-place with matched weights

10. Conclusions

This project is a miniature Physical AI delivery: perception, multimodal policy, closed-loop control, quantitative evaluation, CI, and documentation. For Grid Dynamics it evidences readiness for GAIN-style engagements: digital-twin validation, imitation/VLA integration, KPI-driven demos, and honest sim-to-real scoping without

overclaiming Omniverse fidelity or zero-shot hardware transfer.

11. Reproduce

PowerShell (use the project .venv Python 3.11):

```
.\.venv\Scripts\python.exe main.py --policy act --seed 36
```

```
.\.venv\Scripts\python.exe prompt_ablation.py
```

```
.\.venv\Scripts\python.exe compare_policies.py --seed 36
```

```
.\.venv\Scripts\python.exe evaluate.py --policy act --seeds 0-49 --continue-after-success
```